

BIOS vs UEFI Comparison

Week 3 Portfolio Project
Course: ITEP 414, System Administration and Maintenance
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Section: BSIT 4E

Feature	BIOS	UEFI
Definition	Basic Input/Output System. Firmware built into the motherboard that starts up the hardware and passes control to the operating system.	Unified Extensible Firmware Interface. A newer type of firmware that replaces BIOS and works more like a small operating system itself.
Boot Process	Boots using the Master Boot Record (MBR), which is a small section at the start of the disk that points to the bootloader.	Boots using the GUID Partition Table (GPT) and can load the OS directly from the EFI System Partition.
Disk Support	Can only handle disks up to 2 TB because of how MBR addresses sectors.	Can support disks way bigger than 2 TB since GPT uses 64-bit addressing.
Partition Style	MBR, which only allows up to 4 primary partitions.	GPT, which allows up to 128 partitions.
Security	No real security built in, so it's easier for malware to attack the boot process.	Has Secure Boot, which checks that the bootloader and OS haven't been tampered with before letting the system boot.
Speed	Boots slower since it loads hardware one piece at a time in an old 16-bit mode.	Boots faster because it can initialize hardware at the same time and runs in 32-bit or 64-bit mode.
Advantages	Simple and works on old hardware.	Faster boots, supports big drives, more secure, and has a nicer setup menu you can navigate with a mouse.
Disadvantages	Slow, can't handle large drives, no security features, and the interface is just plain text.	A bit more complex, and old operating systems might not even support it.
Modern Usage	Barely used anymore, mostly just on old machines or legacy systems.	Basically the standard now on all new computers and servers, including the VM used for this project.

Why UEFI Has Mostly Replaced BIOS

After doing this project, it makes a lot more sense why almost every computer today uses UEFI instead of BIOS. BIOS is honestly just outdated technology. It was designed decades ago and it shows, especially with how it handles storage. Since BIOS uses the MBR partitioning system, it can't recognize drives bigger than 2 TB, which is a huge problem now that most storage devices are way past that size. UEFI fixes this by using GPT instead, which supports much bigger drives and way more partitions.

Security is probably the biggest reason UEFI took over though. BIOS has no way of checking if the code it's running during boot is actually safe, so it's an easy target for malware that hides before the OS even loads. UEFI solves this with Secure Boot, which basically checks the digital signature of whatever is trying to boot and blocks it if it looks suspicious or unauthorized. For something like an enterprise server, that extra layer of protection actually matters a lot.

UEFI is also just faster and more convenient to use. BIOS loads hardware step by step in an old 16-bit mode, which slows everything down. UEFI can load things in parallel and run in 32-bit or 64-bit mode, so systems boot quicker. It also has a much better interface. Instead of a plain text menu you navigate with arrow keys, UEFI usually gives you a graphical menu you can click through, which makes troubleshooting or changing settings way less annoying.

Overall, UEFI just does everything BIOS does, but better. It's faster, safer, supports modern hardware properly, and is easier to use. That's why pretty much every new device today, including the Ubuntu Server VM installed for this project, uses UEFI instead of BIOS. BIOS still shows up occasionally on older machines, but it's basically on its way out at this point.